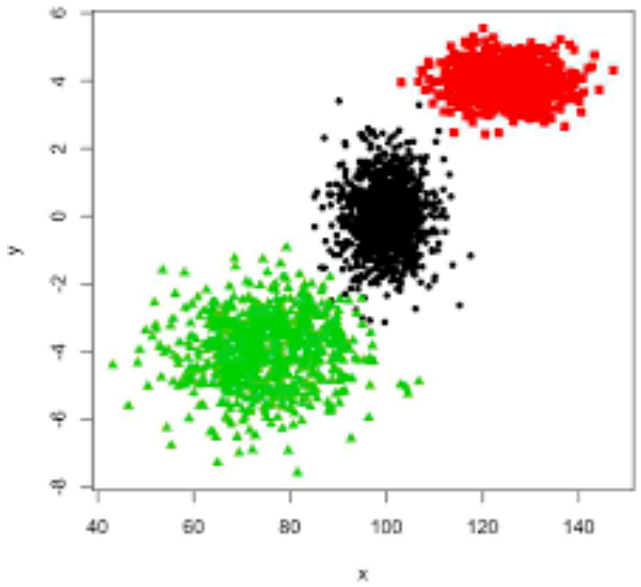
Cluster Analysis

Unsupervised Learning

- Representative problems:
  - Clustering
  - Embeddings



- Clustering: Partition data into similar groups.
- Embeddings: Find good representations of data.

Difference

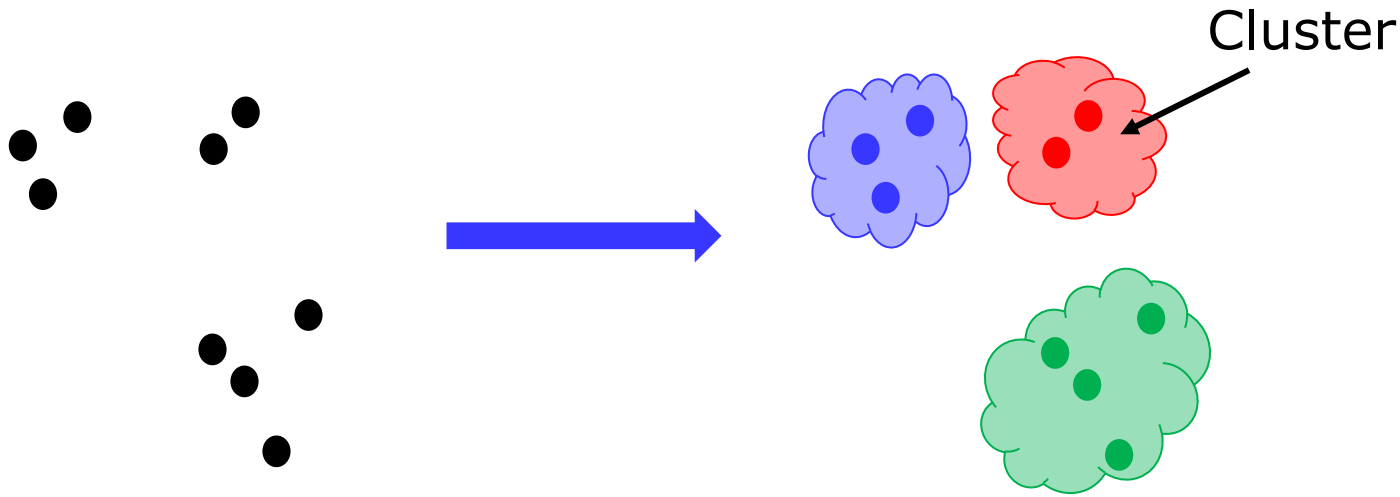
- Supervised Learning:
  - Unknown target $f : R^d \rightarrow Y$
  - Input: $(x_1,f(x_1)),\ldots,(x_n,f(x_n))$
  - Goal: Predict f(x) on new x. Learn f.

- Unsupervised Learning:
  - Input: $x_1,\ldots,x_n$
  - Goal:
    - Group/Cluster/Partition based on similarity in x_i's.
    - Alter x_i's to help learning/speed up algorithms.

- Clustering focus of this lecture

Clustering

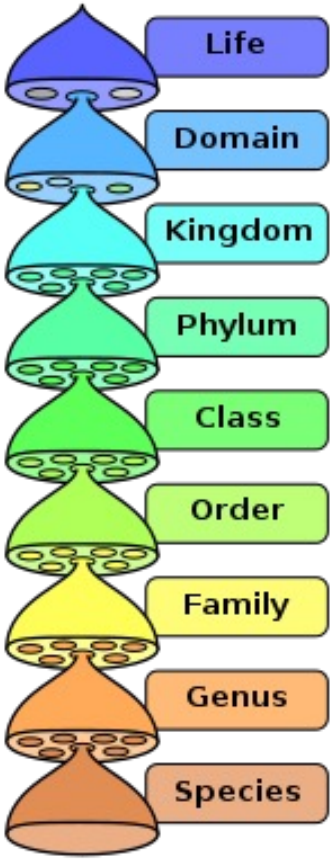
- What is clustering?
  - Partitioning of data into groups/clusters.



- Goals:
  - Clustering for understanding
  - Clustering for utility

Clustering for Understanding

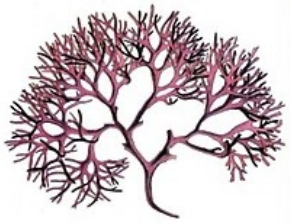
- Clusters
  - Groups of similar items
  - A kind of classification
- Biology Example
- Kingdom
  - Six classes/clusters

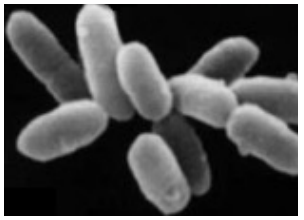


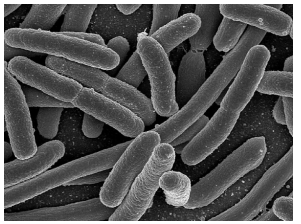

Animalia


Plantae


Fungi


Protista


Archaea


Bacteria

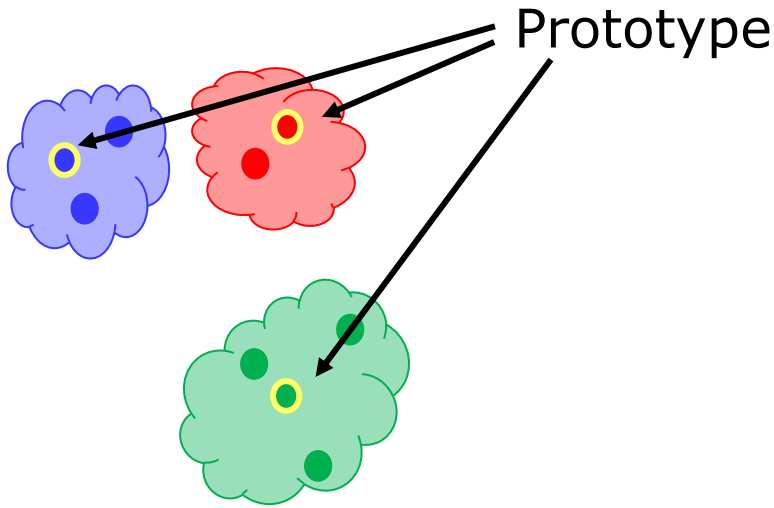
Clustering for Understanding



- Clusters
  - Groups of similar items
  - A kind of classification
- Psychology Example
  - Different kinds of depressions
  - Useful for talking about/treating individual patients
    - Major Depression
    - Persistent Depressive Disorder
    - Bipolar Disorder
    - Seasonal Affective Disorder (SAD)
    - Psychotic Depression
    - …

Clustering for Utility

- Prototypes
  - Many clustering techniques produce a *prototype* for each cluster.
    - Representative of cluster



- Speed up learning algorithms
  - Some algorithms are slow, e.g.
    - Support Vector Machines
    - PCA (next lecture)
  - Train on prototypes instead of full data
    - Small loss in accuracy if good prototypes
    - Great gains in speed



MNIST prototypes

Clustering vs. Classification

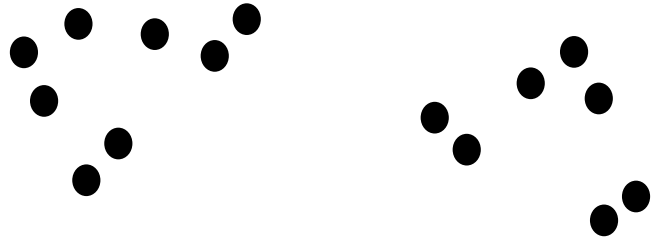
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- Unsupervised Learning:
  - Input: $x_1,\ldots,x_n$
  - Goal:
    - Group/Cluster/Partition based on similarity in x_i's.

- In some sense, clustering is assigning labels to the points.
- Unlike supervised learning, based solely on x_i's, i.e. no labels available during training.

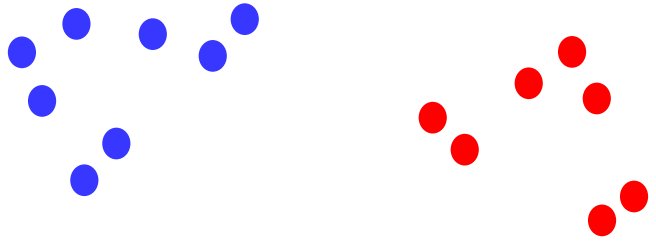
What should the clusters be?

- Cluster Analysis
  - Often not a clear definition
  - Points in same cluster should be similar
  - Points in different clusters should be dissimilar
  - Not always clear how many clusters there should be



What should the clusters be?

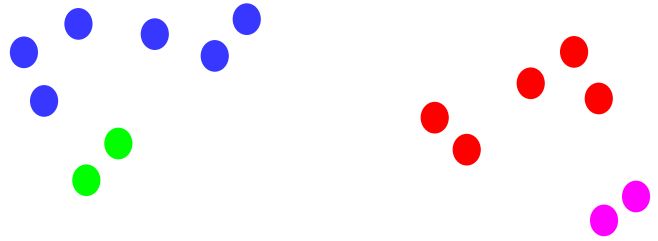
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2 clusters

What should the clusters be?

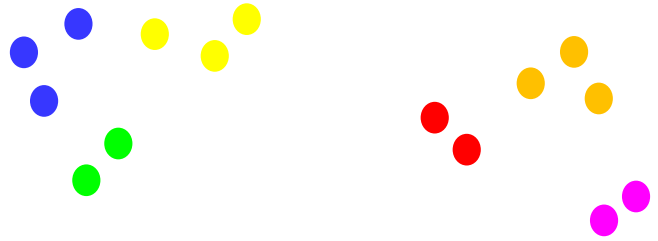
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4 clusters

What should the clusters be?

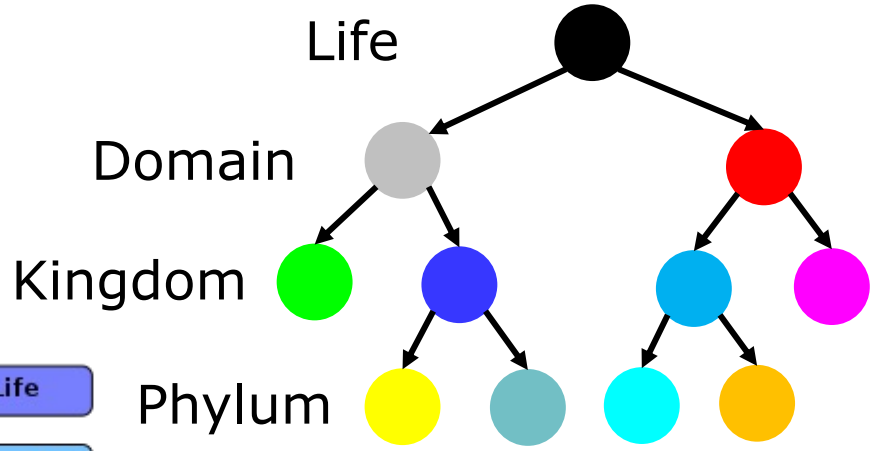
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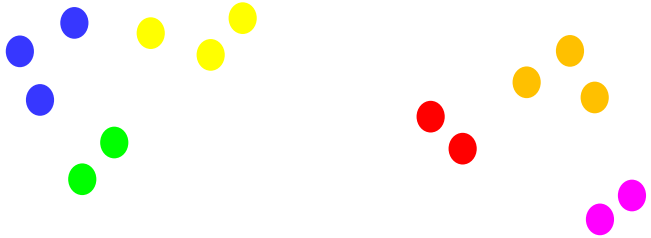


6 clusters

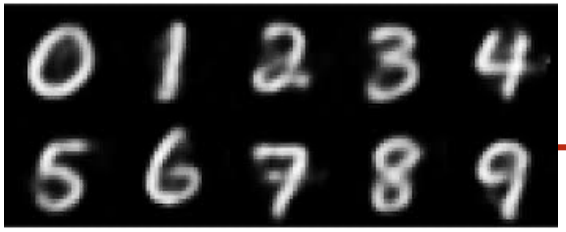
Different types of clusterings

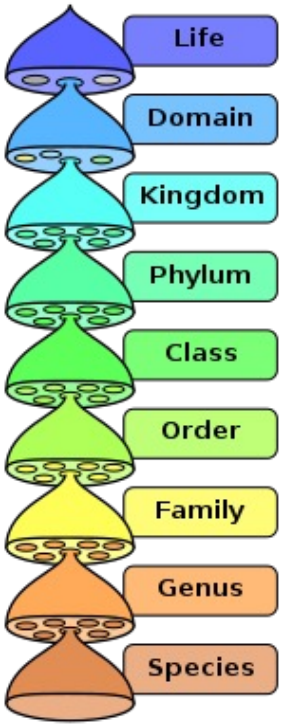
- Hierarchical vs. Partitional
  - Partitional is *flat* like just seen
  - Hierarchical is a *tree* structure

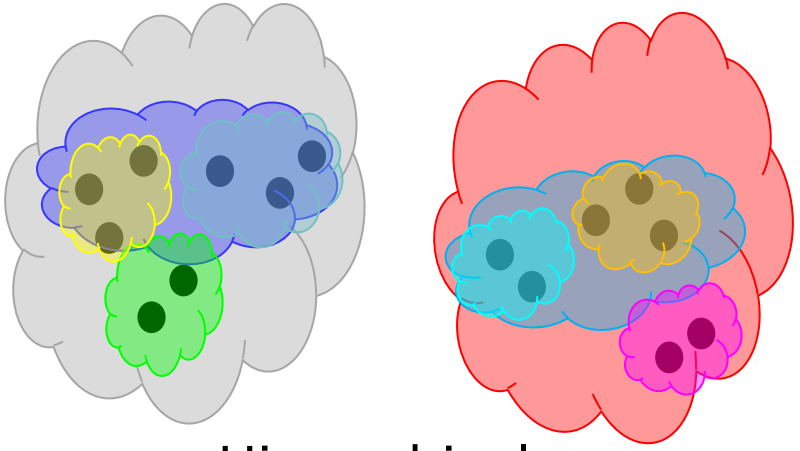




Partitional







Hierarchical

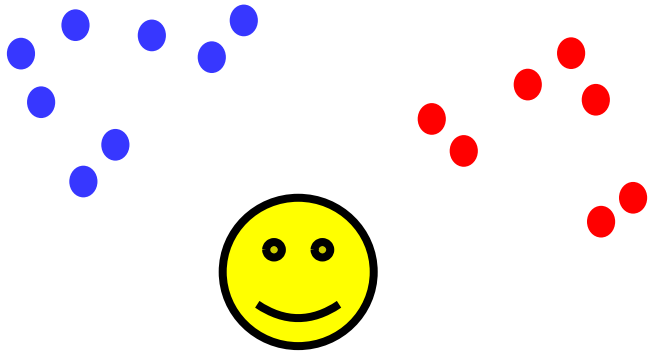
Our Focus

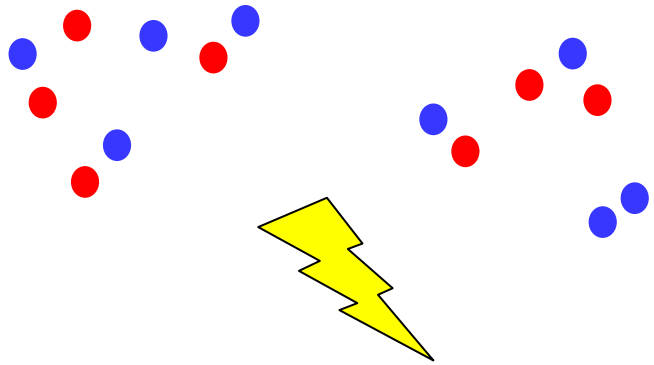
- This Lecture
  - We focus on a concrete clustering approach
  - k-means
    - Partitional
    - Creates k classes
      - k is given by ”user”
      - Assume knows k
      - Alternatively, can cluster with different k and see what you prefer

K-Means Clustering

What should the clusters be?

- Cluster Analysis
  - Points in same cluster should be similar
  - Points in different clusters should be dissimilar

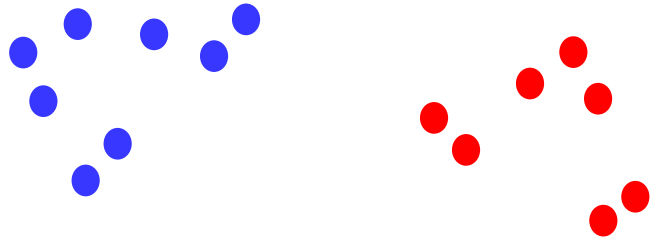




- K-means:
  - Concrete way of choosing how to cluster
  - A measure of how *good* a clustering is, s.t. we prefer left

K-Means Clustering

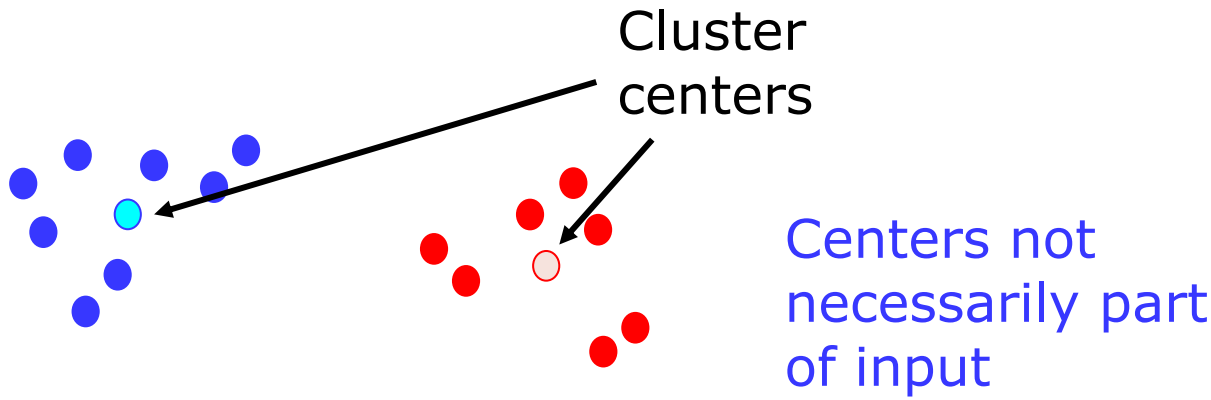
- K-Means
  - Given k: The number of clusters to construct
  - Data is d-dimensional feature vectors/points
    - Best if data scaled s.t. magnitude of coordinates similar



- Prototype based
  - Each of k clusters is represented by a prototype, called the *cluster center*.

K-Means Clustering

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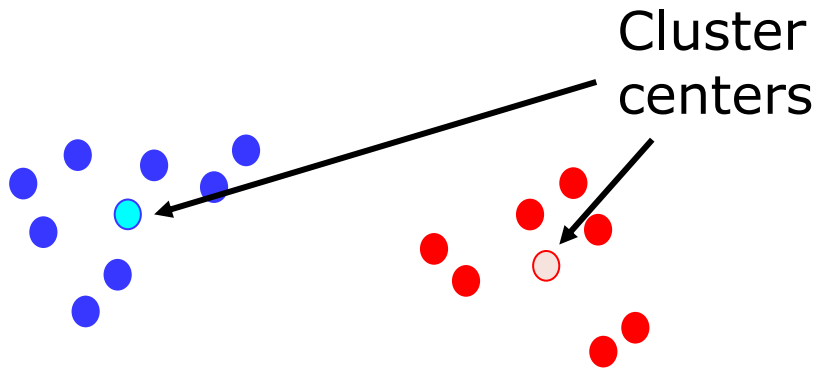
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K-Means Clustering

Recall

$$\|x-y\|^2 = \sum_{i=1}^{d} (x_i - y_i)^2$$

- K-Means
  - Given k: The number of clusters to construct
  - Each of k clusters is represented by a prototype, called the *cluster center*.



- Minimizing a loss:
  - Choose centers C={c_1,…,c_k} to minimize loss on data S:
    - $cost(S,C) = \sum_{x \in S} \min_i \|x - c_i\|^2$
- A kind of least squares loss. A point x pays squared Euclidian distance to nearest center.

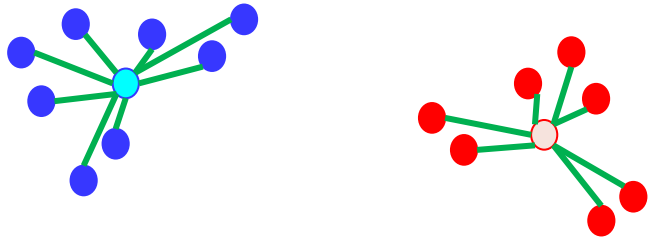
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cost(S,C)=sum squared lengths of green segments



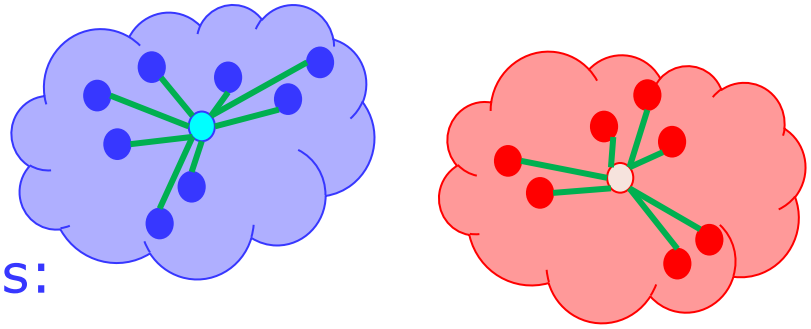
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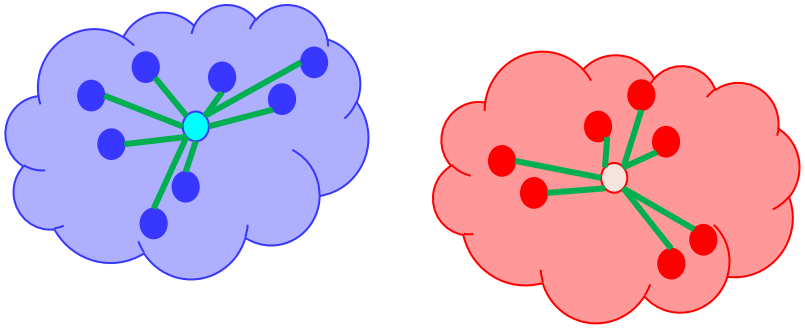
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- Cluster i is those points whose closest center is c_i.

K-Means Clustering

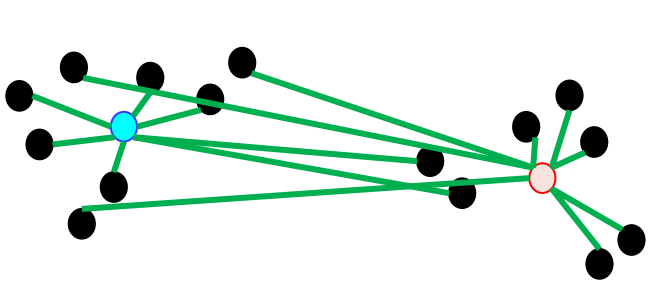
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    - $cost(S, C) = \sum_{x \in S} \min_i \|x - c_i\|^2$



- Solving this optimization problem?
  - NP-hard (even for d=2)
  - Need heuristic

Towards Heuristic

- K-Means
  - Choose centers C={c_1,…,c_k} to minimize loss on data S:
    - $cost(S,C) = \sum_{x \in S} \min_i \|x - c_i\|^2$

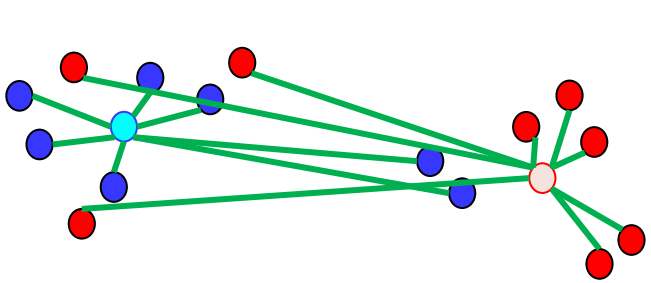


- Let us generalize a bit:
  - For a set of centers C, let $\pi(x) \in C$ denote the cluster center that x is assigned to. Then for assignment π:
    - $cost(S,C,\pi) = \sum_{x \in S} \|x - \pi(x)\|^2$

We allow it to not necessarily be assigned to closest center

Towards Heuristic

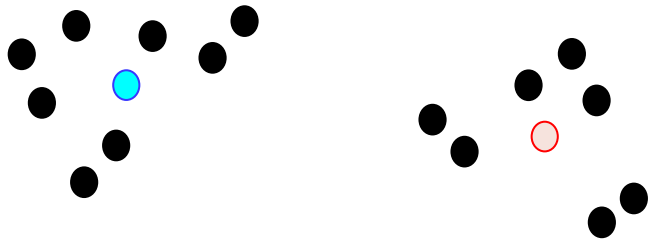
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Towards Heuristic

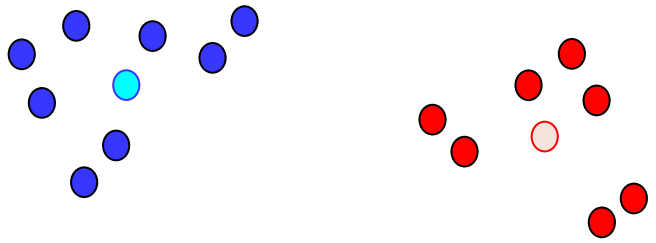
- Easier Subproblem
  - Assume we have already decided on centers $C=\{c_1,\ldots,c_k\}$
  - What assignment π minimizes the cost?
    - $cost(S,C,\pi)=\sum_{x\in S}\|x-\pi(x)\|^2$



- Every point is assigned to closest center!

Towards Heuristic

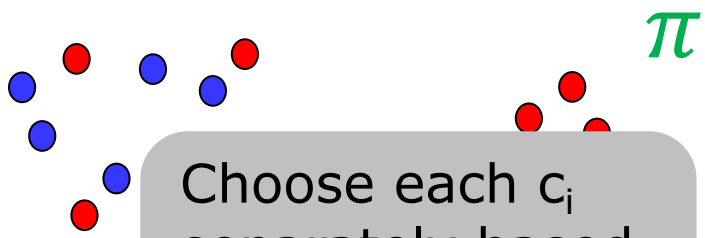
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- Every point is assigned to closest center!
- Ok, that was easy.
  - What else can we say?

Towards Heuristic

- Easier Subproblem
  - Assume we have already decided on assignment π
    - i.e. which points that go together in each cluster
  - What centers C={c_1,…,c_k} minimizes the cost?
    - $cost(S, C, \pi) = \sum_{x \in S} \|x - \pi(x)\|^2$



Choose each c_i separately based on points x with $\pi(x) = c_i$

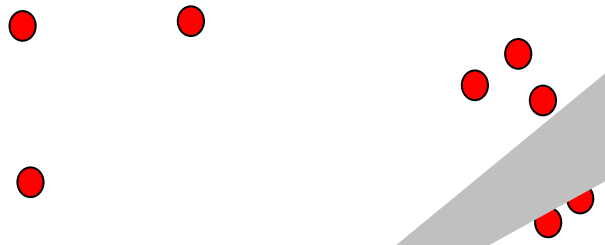
Sum of contributions from individual clusters

- Rewrite cost:
  - $cost(S, C, \pi) = \sum_{x \in S} \|x - \pi(x)\|^2 = \sum_{c_i \in C} \sum_{x \in S: \pi(x) = c_i} \|x - c_i\|^2$

Contribution from cluster i depends only on c_i.

Towards Heuristic

- Easier Subproblem
  - Assume we have already decided which points S_i that belong to cluster i.
  - What center c_i minimizes the cost?
    - $cost(S_i, c_i) = \sum_{x \in S_i} \|x - c_i\|^2$



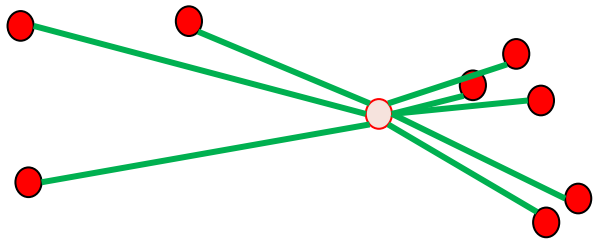
To minimize sum of squares, coordinate j should be chosen as the mean $c_i(j) = \frac{1}{|S_i|} \sum_{x \in S_i} x(j)$

  - Expanding cost:
    - $cost(S_i, c_i) = \sum_{x \in S_i} \|x - c_i\|^2 = \sum_{x \in S_i} \sum_j \big(x(j) - c_i(j)\big)^2 = \sum_j \sum_{x \in S_i} \big(x(j) - c_i(j)\big)^2$

Independent across coordinates

Towards Heuristic

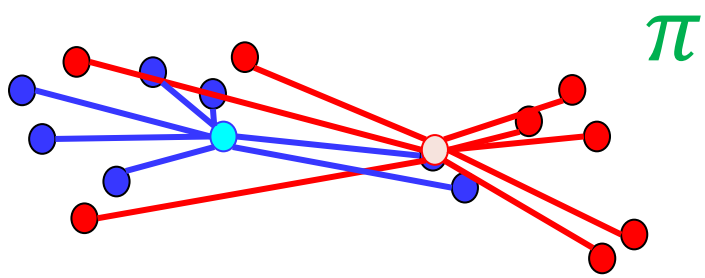
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  - What center c_i minimizes the cost?
    - $cost(S_i, c_i) = \sum_{x \in S_i} \|x - c_i\|^2$



  - The mean!
    - $c_i = \frac{1}{|S_i|} \sum_{x \in S_i} x$
  - Hence the name k-means!

Towards Heuristic

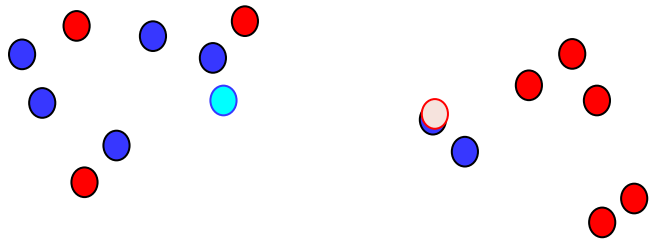
- Easier Subproblem
  - Assume we have already decided on assignment π
    - i.e. which points that go together in each cluster
  - What centers C={c_1,…,c_k} minimizes the cost?
    - $cost(S, C, \pi) = \sum_{x \in S} \|x - \pi(x)\|^2$



- The means of each cluster!
  - Observation: We can easily improve this clustering!

Towards Heuristic

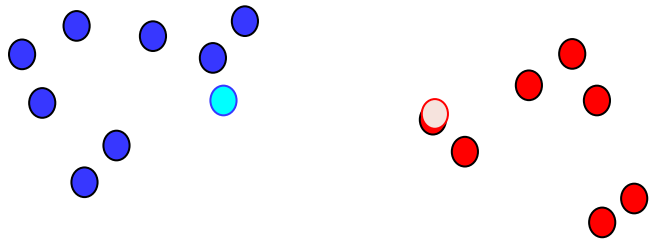
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  - Assume we have already decided on centers $C=\{c_1,\ldots,c_k\}$
  - What assignment π minimizes the cost?
    - $cost(S,C,\pi)=\sum_{x\in S}\|x-\pi(x)\|^2$



- Every point is assigned to closest center!

Towards Heuristic

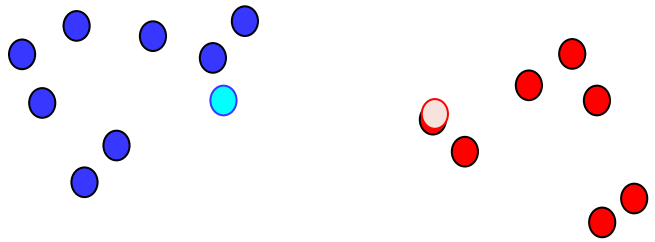
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  - Observation: We can easily improve this clustering!

Towards Heuristic

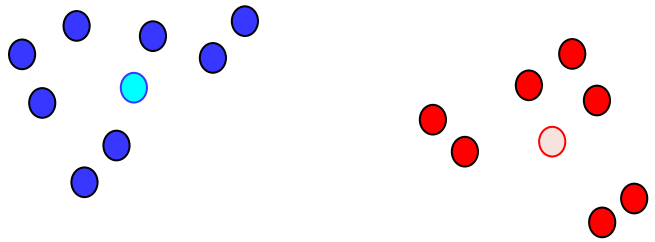
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  - Assume we have already decided on assignment π
    - i.e. which points that go together in each cluster
  - What centers C={c_1,…,c_k} minimizes the cost?
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- The means of each cluster!

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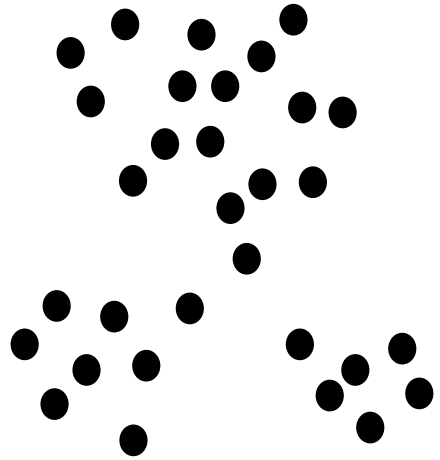
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- The means of each cluster!
  - Observation: Every point still closest to its assigned center.
    - The process converged

Lloyd's Algorithm

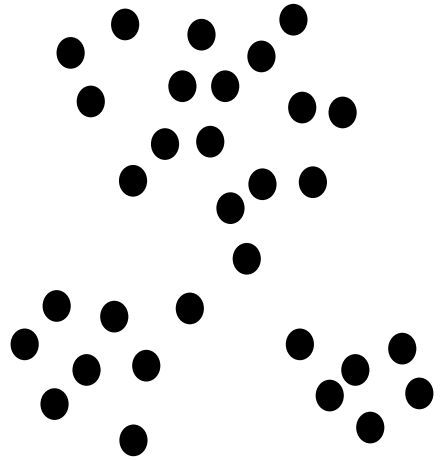
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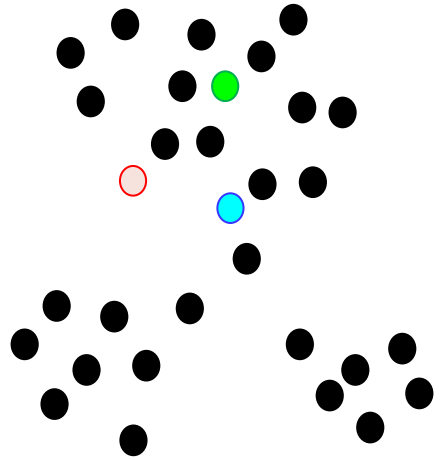
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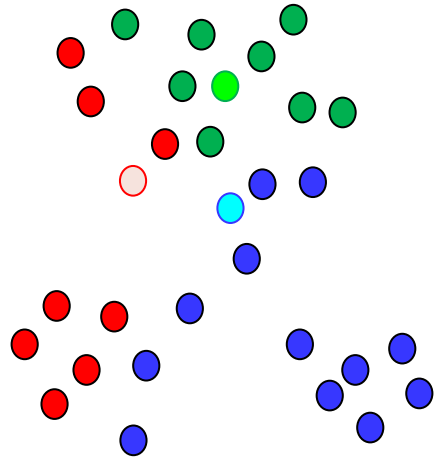
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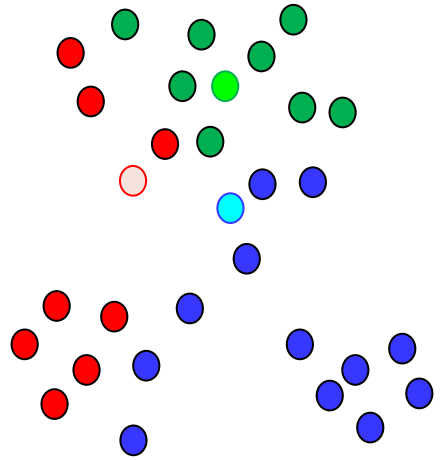
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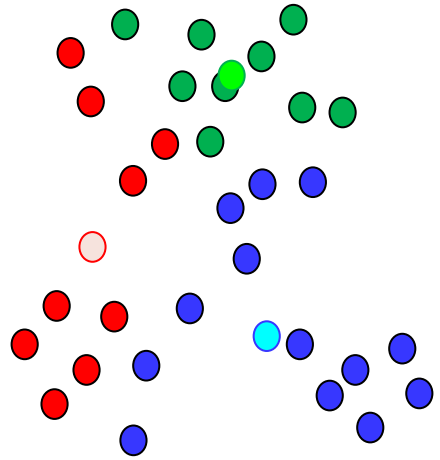
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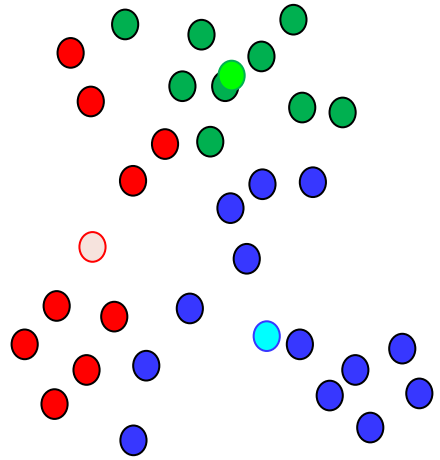
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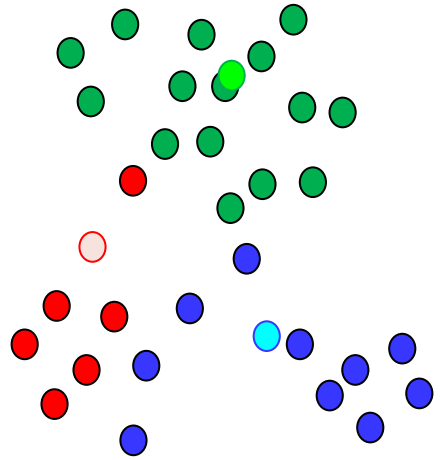
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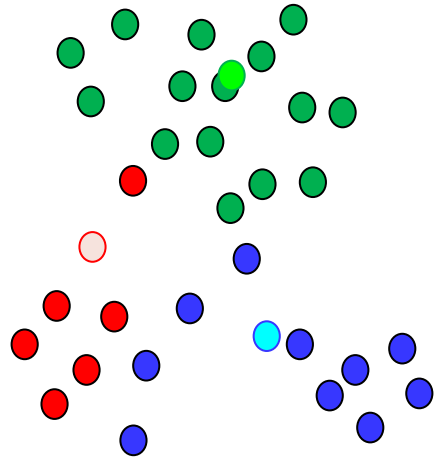
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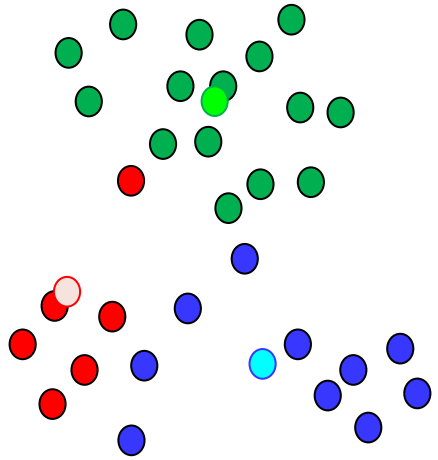
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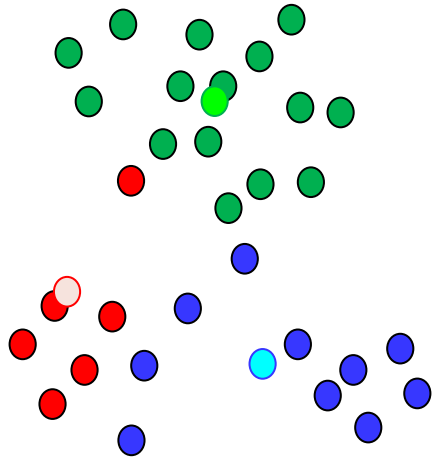
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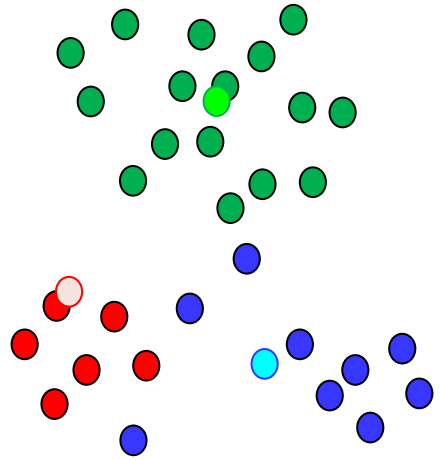
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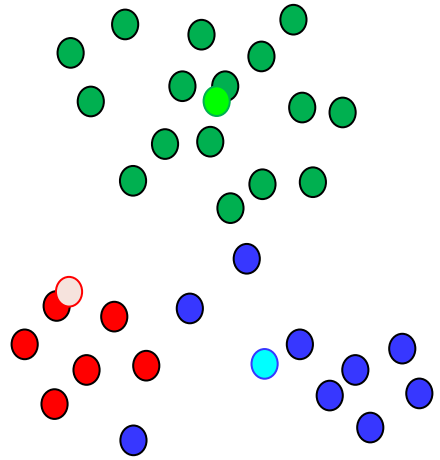
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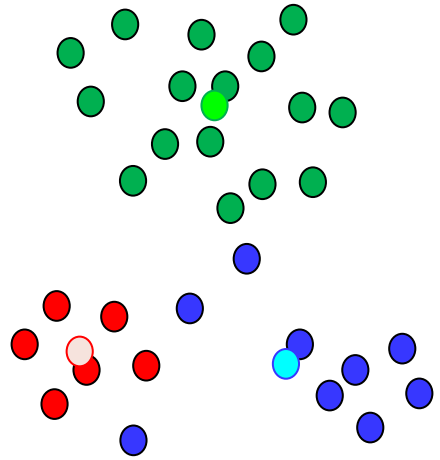
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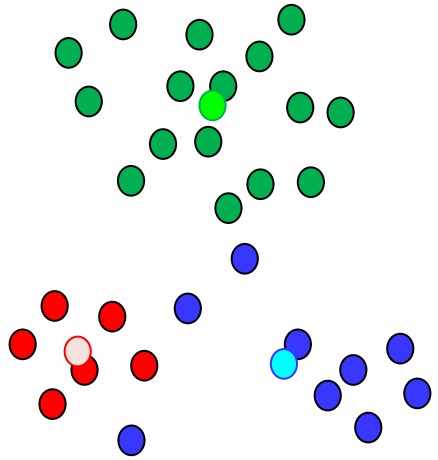
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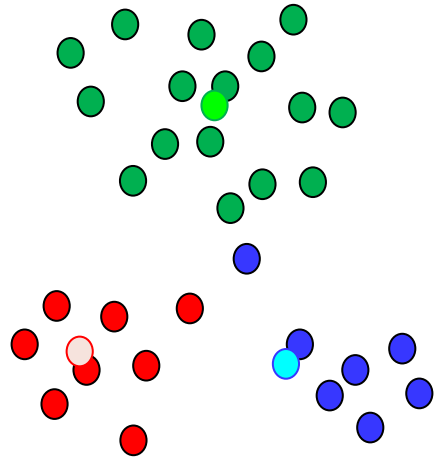
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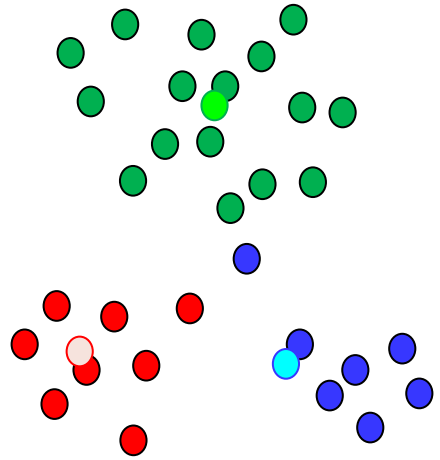
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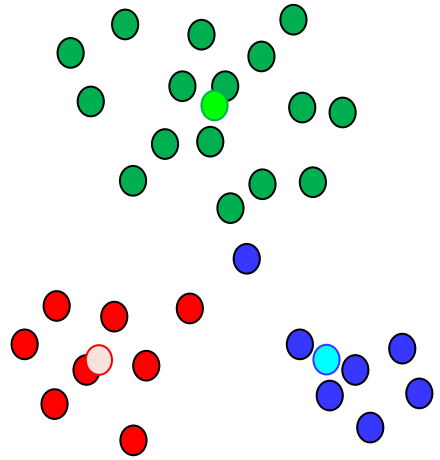
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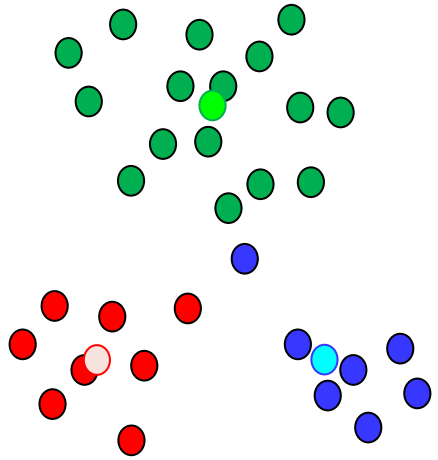
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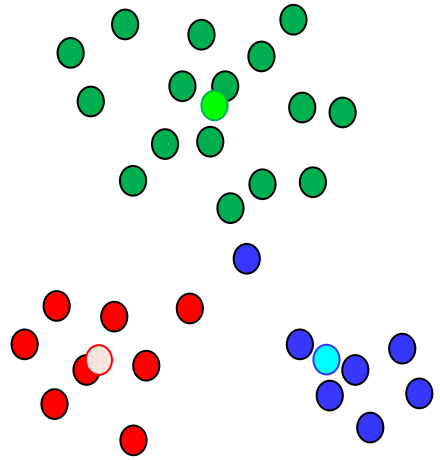
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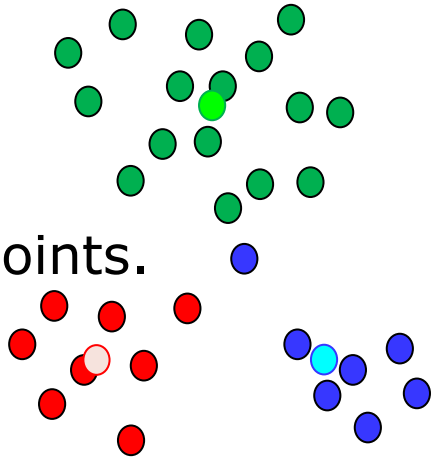
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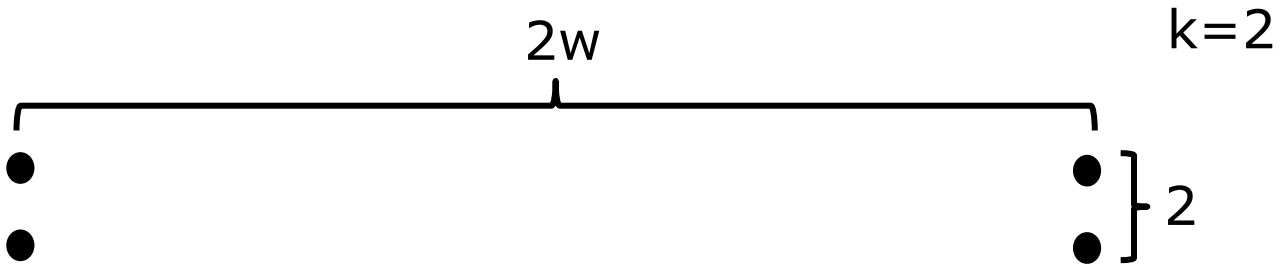


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- Properties:
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  - Quality of final clustering depends on initialization
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Lloyd's Algorithm

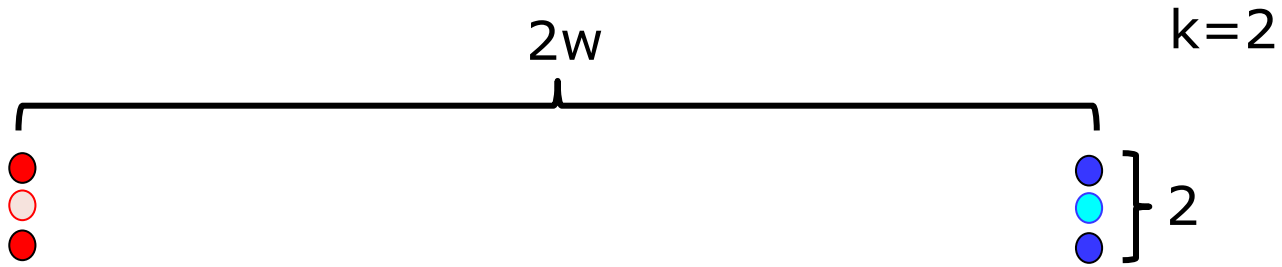
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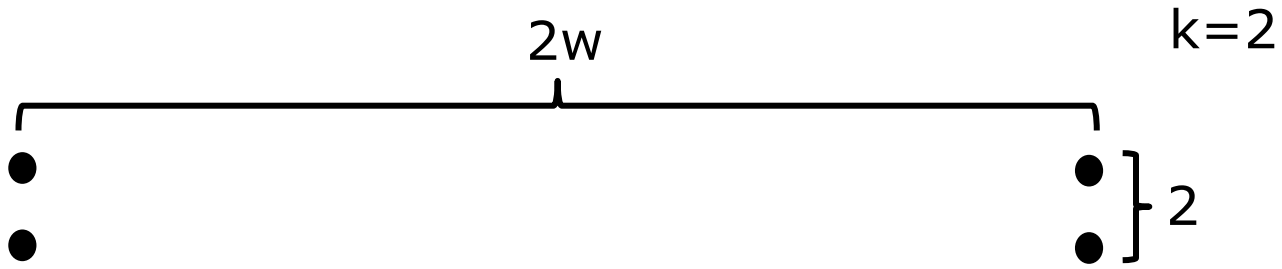


Optimal clustering C* has cost $cost(S, C^*) = \sum_{x \in S} \min_i \|x - c_i^*\|^2 = 4$

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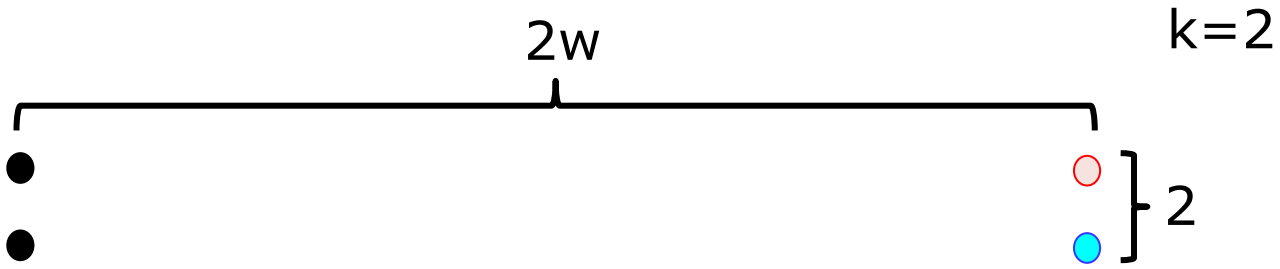


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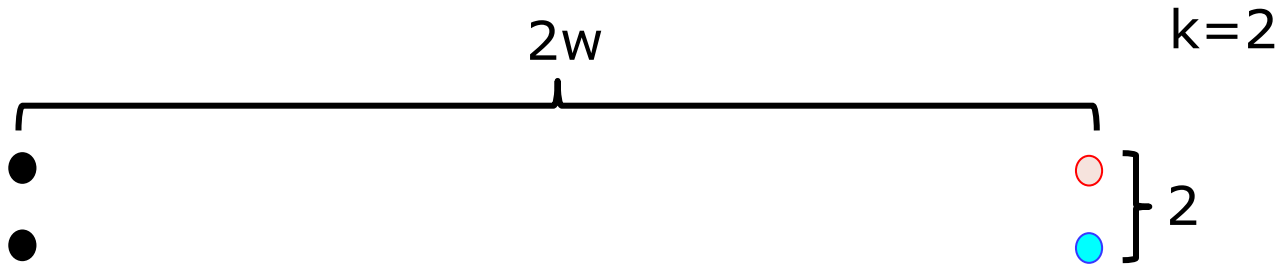


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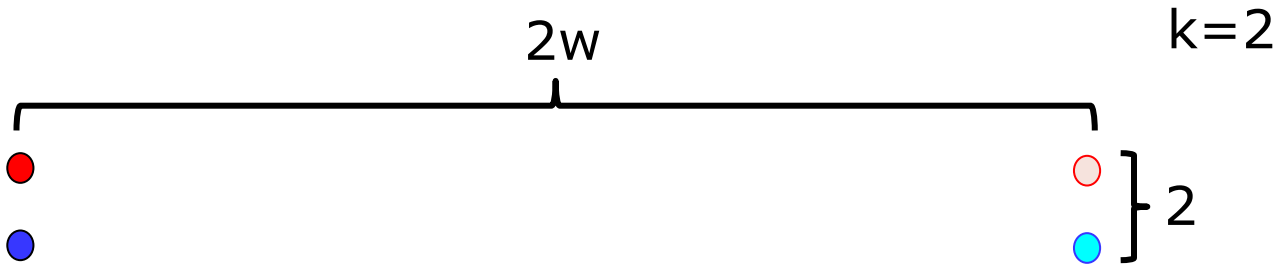


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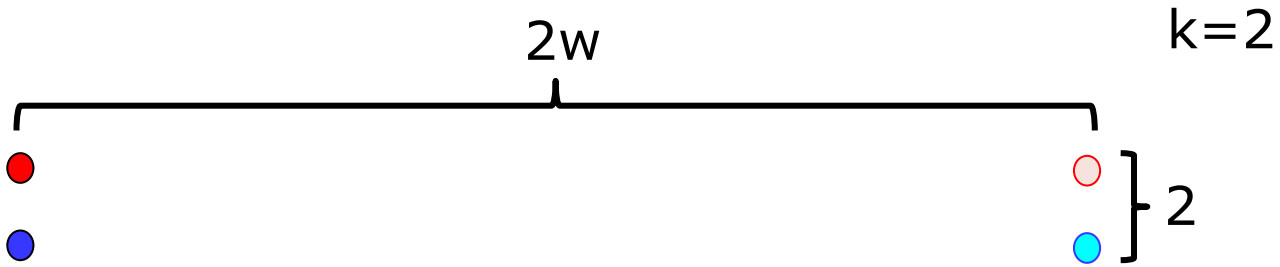


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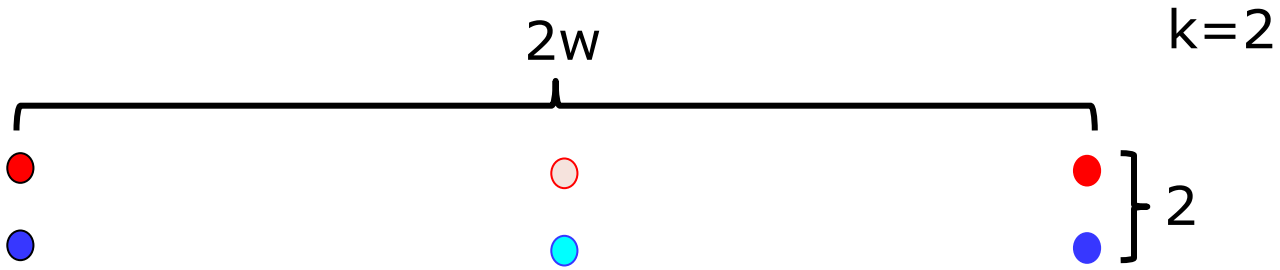


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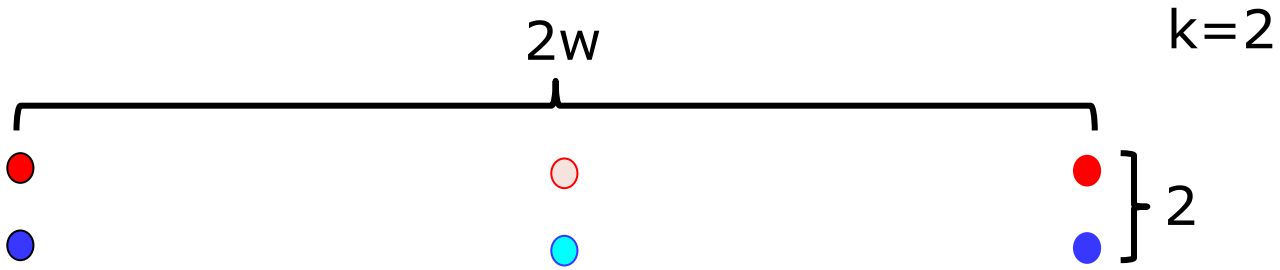


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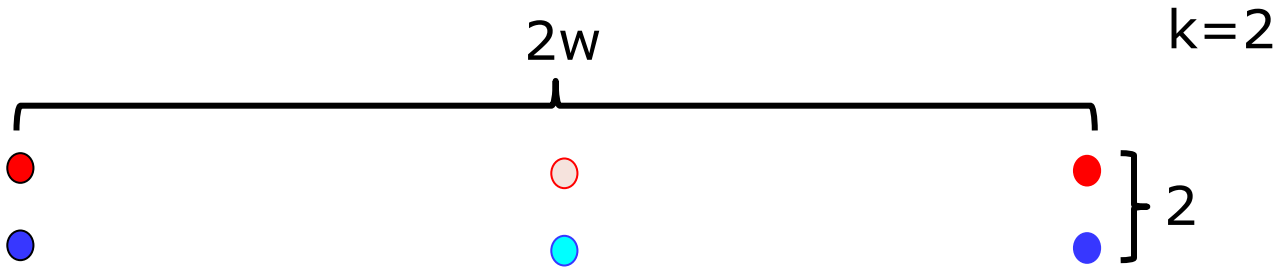


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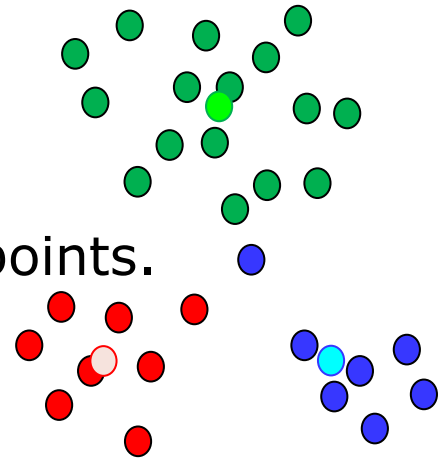


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Found clustering C has cost $cost(S, C) = \sum_{x \in S} \min_i \|x - c_i\|^2 = 4w^2$

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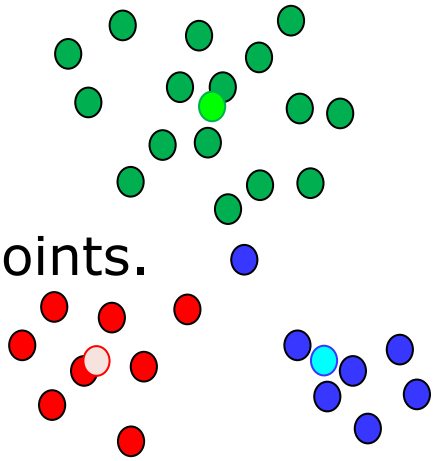


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- Properties:
  - Always converges since each step reduces cost
  - Quality of final clustering depends on initialization
    - May be arbitrarily worse than optimal clustering
  - May take exponential number of steps to converge
    - Terminate if improvement in cost less than some ε-fraction

Typically not in practice

Lloyd's Algorithm



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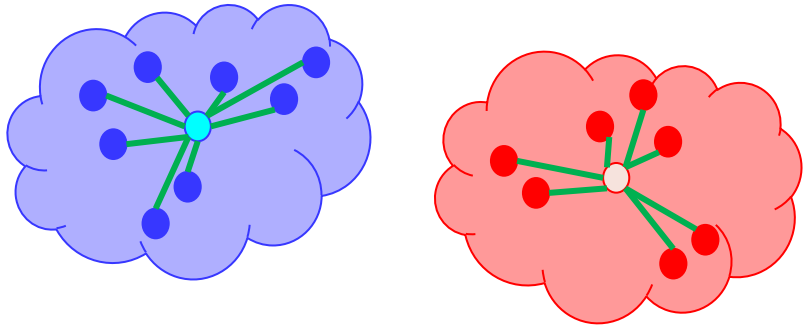
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  - Quality of final clustering depends on initialization
    - May be arbitrarily worse than optimal clustering
  - May take exponential number of steps to converge
    - Terminate if improvement in cost less than some ε-fraction

Typically not in practice

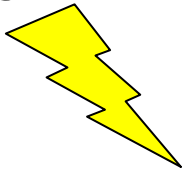
K-Means++

K-Means Clustering

- K-Means
  - Choose centers C={c_1,…,c_k} to minimize loss on data S:
    - $cost(S, C) = \sum_{x \in S} \min_i \|x - c_i\|^2$

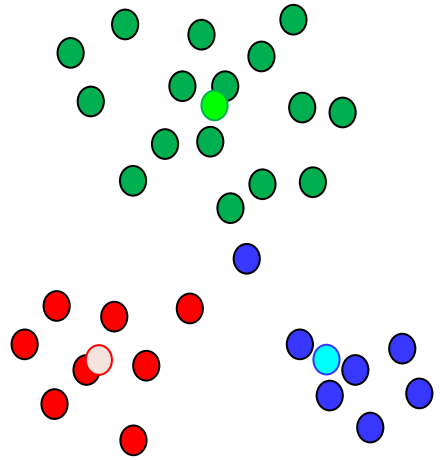


- Solving this optimization problem?
  - NP-hard (even for d=2)
  - Need heuristic



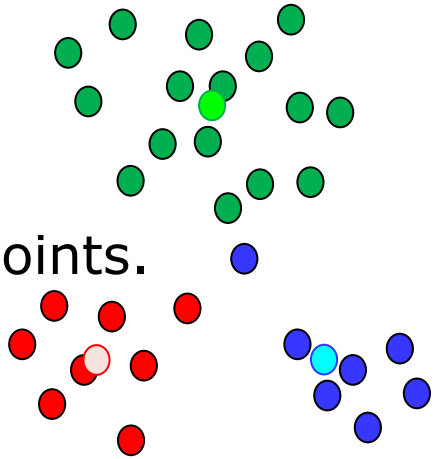
Lloyd's Algorithm

- Most famous k-means algorithm/heuristic
  - Often called the k-means algorithm



- Lloyd's algorithm:
  - Pick k random input points and let C equal those points.
  - Repeat until convergence:
    - Assign every point to its closest center in C
    - Replace C by the means of each cluster

Lloyd's Algorithm

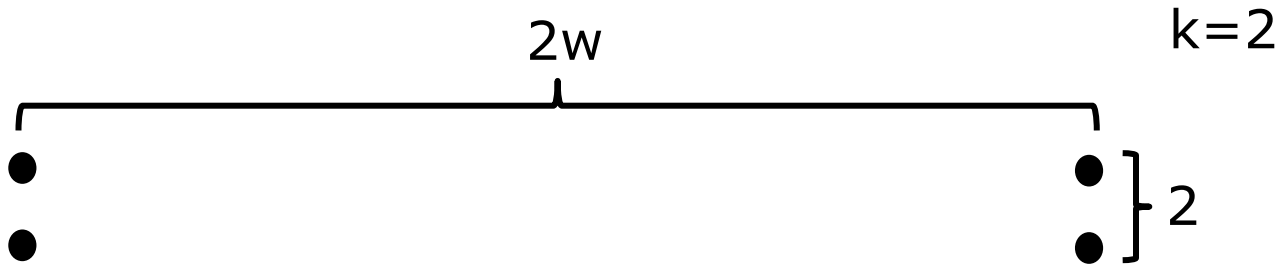


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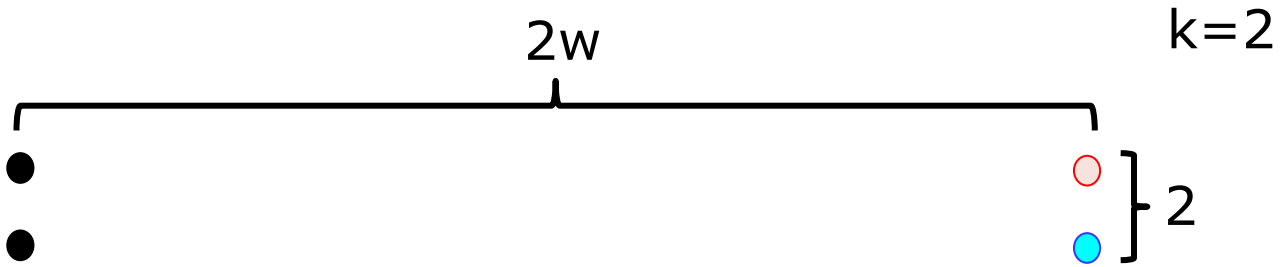
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- Obtained clustering may be arbitrarily bad!

Lloyd's Algorithm

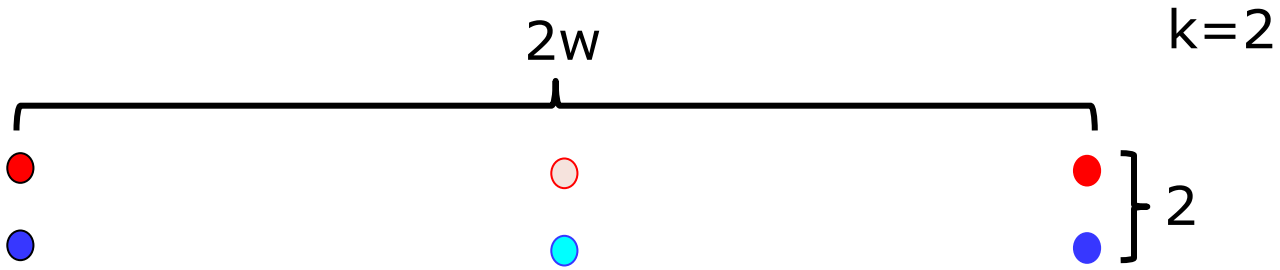
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Optimal clustering C* has cost $cost(S, C^*) = \sum_{x \in S} \min_i \|x - c_i^*\|^2 = 4$

Found clustering C has cost $cost(S, C) = \sum_{x \in S} \min_i \|x - c_i\|^2 = 4w^2$

- Obtained clustering may be arbitrarily bad!

Lloyd's Algorithm

- Lloyd's algorithm:
  - Pick k random input points and let C equal those points.
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- K-means++ algorithm:
  - Better initialization strategy
  - Then Lloyd's

Motivating k-means++

- Let us look at bad example for Lloyd's again:



- Think of picking the initial centers one at a time among input points.
  - Look at cost along the way

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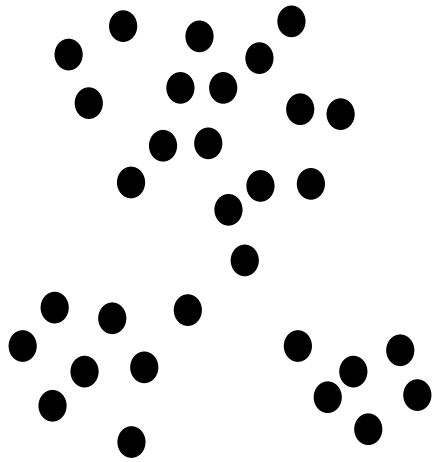
- Let us look at bad example for Lloyd's again:



- Think of picking the initial centers one at a time among input points.
  - Look at cost along the way
  - Seems like a bad idea to choose a center so close to first center!
  - Other two points still very high costs!

Motivating k-means++

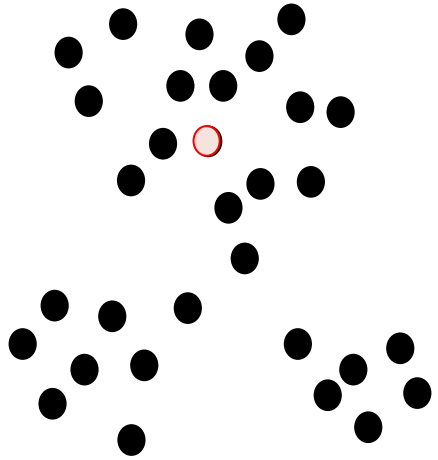
- General principle



- Think of picking the initial centers one at a time among input points.
  - Look at cost along the way
  - Seems like a bad idea to choose a center close to current centers

Motivating k-means++

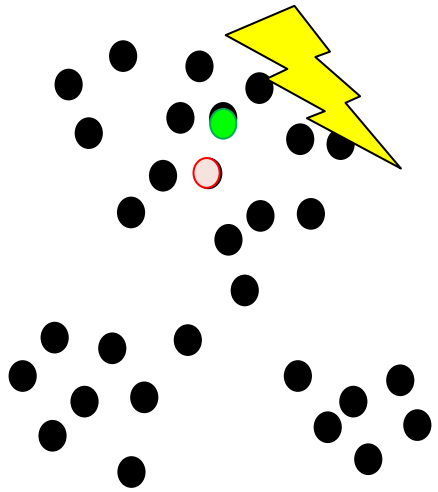
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Motivating k-means++

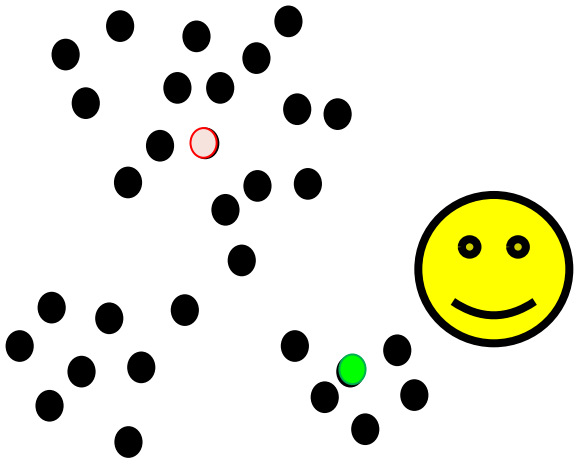
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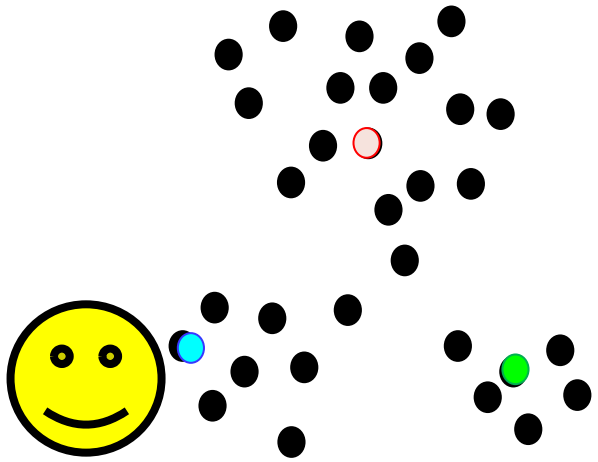
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K-Means++ Algorithm

- K-means++ algorithm:
  - Let c_1 be a uniform random input point.
  - For i=2,…,k:
    - Pick c_i to equal input point x with probability
      - $p_x = \frac{\min_{j<i}\|x-c_j\|^2}{\sum_{y\in S}\min_{j<i}\|y-c_j\|^2}$

p_i proportional to contribution to cost of clustering with $c_1,\ldots,c_{i-1}$

  - Repeat until convergence:
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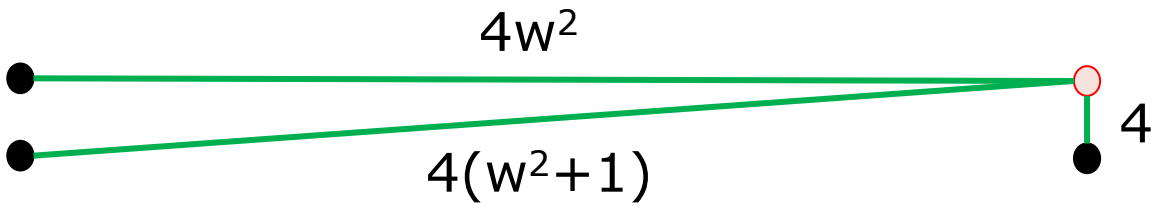
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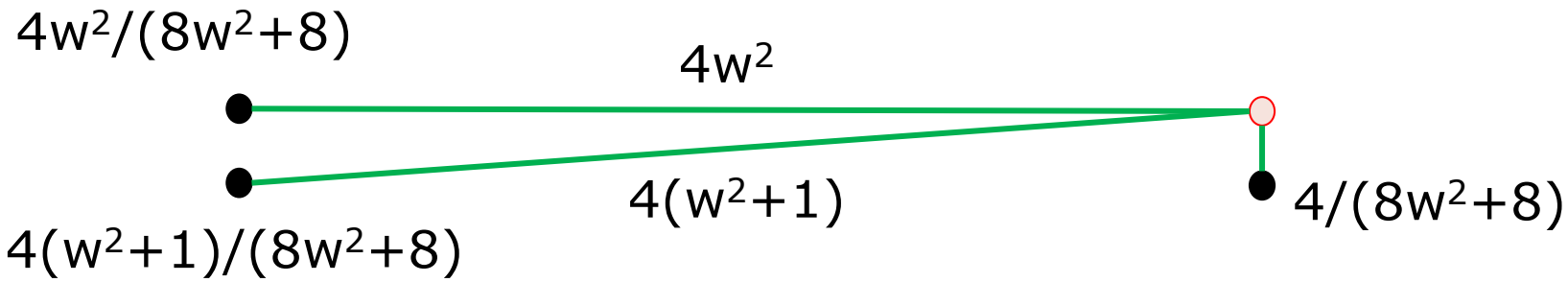
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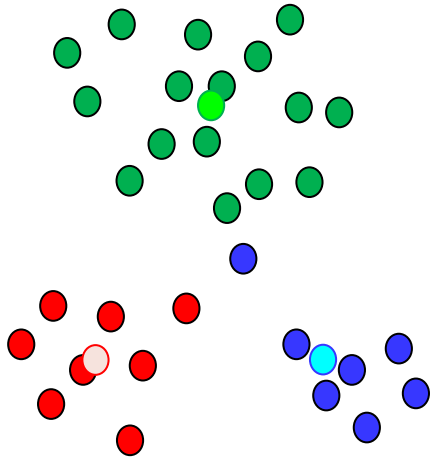


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- Lloyd's algorithm with clever initialization.
  - Can prove: expected cost of solution O(logk OPT).
    - OPT=cost of optimal clustering